CORRECTION TO THE 11-07 TWIN NOTE: THE SHAPE-LEVEL HUNCH FAILS IN S_{16} (AND WHAT S_{15} SHOWED)

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Corresponds to work-in-progress commit `498fdd1` (`grandpa-rick/work-in-progress`; data in `data/2026-11-07-q68-s16/`).

Corrects: `notes/2026-11-07-q68-kl-cone-twins.tex` (built against `fd667bf`, PDF `812e7b5`), §6 item 1 (source line 258).

Sources: `proofs/2026-11-07-q68-s15-separator.md` (PROVE 281), `proofs/wake282/sieve-results.md`, `proofs/wake282/verify_indep/`; registry `proofs/registry/seedQ2-kl-cone.json`.

The correction in one paragraph. The 11-07 note stated, as a hunch (§6, item 1; also the third-from-last line of its claims box): *"in type A, a twin exists in shape λ iff $(\star)$ fails, i.e. iff $s_\lambda = \sum F_{\mathrm{Des}T}$ contains F_A, F_B with $\varnothing \neq B \subseteq \operatorname{int} A$."* **This is false at $n = 16$.** The shapes $(5,3,3,3,2)$, $(4,4,3,3,2)$, $(5,5,4,2)$ and $(5,5,4,1,1)$ carry $(\star)$-violations but have no dominated twin (computed, independently replayed; §3). It survived $n = 14$ (4/4) and $n = 15$ (9/9 shapes), and fails at $n = 16$ (17/21 shapes). **Everything else in the 11-07 note is unaffected:** Lemma A, Theorem B, C1, C2, C4, Theorem C3, Lemma S, Theorems T14 and T14′ are proved statements that do not depend on the hunch, and the computed items there (all-cells check at $n = 14$, Coxeter scans, GW calibration) are unchanged. Only the forward-looking §6 item 1 is withdrawn.

Claims and registry grades (from `seedQ2-kl-cone.json`; nothing upgraded).

Shape-level hunch of the 11-07 note, §6.1	dead-end (refuted, computed)
(λ, A)-level form (every violation is a twin), §2	dead-end (refuted at $n = 15$)
Lemma P (parity), §2	proved
Proposition U (below-x obstruction, $n = 15$), §2	proved (certificate)
S_{15} census: 48 pairs, 26 twins, §2	proved (computer-assisted)
S_{16} census: 410 pairs, 112 TWIN / 291 NO / 7 undecided, §3	computed
Four S_{16} shapes with a violation and no twin, §3	computed
Sieve rule R1 (parity + no (ii)-element) $\Leftrightarrow$ twin, §4	dead-end (48/48 $\to$ 317/403)
(ii)-elements are candidate x-halves (58/58), §4	computed
(ii) always kills (56/56 decided), §4	computed; general: hunch

Grades throughout: hunch < sketched < computed < checked-sober < proved. Conventions are those of the 11-07 note: GW convention C, left cells $\{Q = \text{const}\}$, $L(w) = \mathrm{Des}P(w)$, facts (F1)–(F3), the dominated-twin definition, Lemma A, C1, C2, and Lemma S.

1. Setup recalled

A *candidate pair* in shape $\lambda \vdash n$ is $(P_x, P_z) \in SYT(\lambda)^2$ with $\mathrm{Des}P_x = A$ and $\varnothing \neq \mathrm{Des}P_z \subseteq \operatorname{int} A$. By C2 (proved), every dominated twin (x, z) in a cell of shape λ has $(P(x), P(z))$ a candidate pair. A $(\star)$-violation is a (λ, A) carrying at least one candidate pair. To decide a candidate pair, Lemma A's hypothesis has to be checked at every $y \in \Gamma$ with $L(y) \not\subseteq A$ and

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$\mu(y,x) \neq 0$. By (F1) these are (PROVE 280 Step 2, re-derived in PROVE 281 without the row-count shortcut):

(i) the *star images* $y = sx \in \Gamma$ with $L(sx) \not\subseteq A$; these have $\mu(sx,x) = 1$;
(ii) the $y = (U,Q)$ with $\mathrm{Des}U \supsetneq A$ and $\mu(y,x) \neq 0$ (not required to be below x).

A pair is a twin iff $\mu(y,z) \geq \mu(y,x)$ for all y in (i)∪(ii). Each needed μ is decided by a *certificate*: a path of simultaneous left (or right) star operations, constant in $\tilde{\mu}$ by Lemma S, ending at a Bruhat-comparable pair with length gap 1 ($\mu = 1$), or at a pair with even gap, Bruhat-incomparable, or violating (F1) ($\mu = 0$).

2. What $n = 15$ showed (PROVE 281)

The 11-07 note named $n = 15$ as the separator. Its result, which I had not yet reported to you in this form:

Lemma P (proved)**.** *Let W be simply laced with connected Coxeter graph, and let (x,z) be a dominated twin. Then $\ell(x) \equiv \ell(z) \pmod 2$.*

Proof. By C2, $\varnothing \neq L(z) \subseteq L(x)$, so $L(x) \neq \varnothing$. Also $L(x) \neq S$, since w_0 is a singleton cell and $x \neq z$. The graph is connected, so there are adjacent $s \in L(x)$, $t \notin L(x)$, with $m(s,t) = 3$. Put $x' = *x$. By (F2), $x' \in \Gamma$; also $\mu(x',x) = 1$ and $t \in L(x') \setminus L(x)$. The twin condition gives $\mu(x',z) \geq 1$. A nonzero μ needs an odd length gap, and $\ell(x') = \ell(x) \pm 1$. Hence $\ell(z) \equiv \ell(x)$. □

In type A the parity of $\ell(x) - \ell(z)$ for $x = (P_x, Q)$, $z = (P_z, Q)$ does not depend on Q: a right star changes each length by ± 1, and right stars connect all cells of a shape (checked exhaustively for the $n = 15$ shapes; in general Haiman's dual-equivalence theorem, recall-only). So Lemma P is a condition on (P_x, P_z). At $n = 14$ it held automatically, which is why 4/4 looked like evidence.

Proposition U (proved, certificate)**.** *Let $\lambda = (5,4,3,3)$, $A = \{2,3,4,6,7,8,10,11,12\}$, and*

$$P_x = 1\,2\,6\,10\,15/3\,7\,11\,14/4\,8\,12/5\,9\,13, \qquad P_z = 1\,2\,3\,7\,15/4\,5\,6\,11/8\,9\,10/12\,13\,14,$$
$$P_y = 1\,2\,6\,10\,14/3\,7\,11\,15/4\,8\,12/5\,9\,13,$$

with $\mathrm{Des}P_z = \{3,7,11\} = \mathrm{int}\,A$ and $\mathrm{Des}P_y = A \cup \{14\}$. Then (x,z) passes parity and every star image satisfies $\mu(sx,z) = 1$, yet (x,z) is not a twin in any cell: $\mu(y,x) = 1$ and $\mu(y,z) = 0$.

In the superstandard cell, $\ell(y) = 52 < \ell(x) = 59 < \ell(z) = 77$. $\mu(y,x) = 1$ by the left-star path $(1,2,3,5,6,7,12)$ to a gap-1 Bruhat pair. $\mu(y,z) = 0$ by the right-star path $(4,3,2,1,5,4,3,6,5,7)$ to an incomparable pair; right stars preserve μ by Lemma S applied to inverses, using $P_{a,b} = P_{a^{-1},b^{-1}}$ (Hecke anti-involution $T_w \mapsto T_{w^{-1}}$, one line, KL basics recall-only). Note that P_y is itself the x-half of an S_{15} twin: one twin lift sits next to another and kills it.

Census (proved, computer-assisted). At $n = 15$ there are 34 exact ($\star$)-violations (λ, A) in 9 shapes, namely $\{\lambda + \square : \lambda \text{ an } S_{14} \text{ twin shape}\}$, carrying 48 candidate pairs. Of these, 26 are twins; 20 fail Lemma P and 2 (one automorphism orbit) fail by Prop. U. Ten (λ, A) carry no twin. So the (λ, A)-level form ("every violation is a twin") is **dead at** $n = 15$ (dead-end, refutation proved). The shape-level form, the one stated in the 11-07 note, survived: every one of the 9 shapes has a twin. Verdicts were replayed 48/48 by an independent checker, validated against full KL in S_5, S_6 (0 wrong), and found consistent with $w \mapsto w_0 w w_0$ (26 orbits) and in 53/53 random cells.

3. $n = 16$: THE SHAPE-LEVEL HUNCH FAILS

Census (computed). S_{16} has 227 (λ, A) violations in 21 shapes, carrying 410 candidate pairs. The certificate search returns 112 TWIN, 291 NO, 7 UNDECIDED (star orbit exhausted before a decidable endpoint). A replay checker re-verifies all 410 verdicts. The 7 undecided pairs lie in $(5,5,3,3)$ and $(6,4,3,3)$, both of which have certified twins, so they do not affect the shape count. 17 of the 21 shapes have a twin. The remaining four:

λ	f^λ	NO by parity	$\ell(x), \ell(z)$ in Q_0	kill of the parity-OK pair
$(5,3,3,3,2)$	280280	2 of 3	63, 83	$s_1 x \not\sim_{\text{Bruhat}} z$ **(Bruhat only)**
$(4,4,3,3,2)$	231660	2 of 3	67, 87	$s_1 x \not\sim_{\text{Bruhat}} z$ **(Bruhat only)**
$(5,5,4,2)$	231660	2 of 3	58, 78	$s_{4,8,11}x$: star paths $5,4,6$ (Lemma S)
$(5,5,4,1,1)$	280280	2 of 3	59, 79	$s_{4,8,11}x$: star paths $5,4,6$ (Lemma S)

The four shapes form two conjugate pairs: $(5,3,3,3,2)' = (5,5,4,1,1)$ and $(4,4,3,3,2)' = (5,5,4,2)$. Each has one candidate pair per (λ, A), three A's each. The parity-OK pairs are, rows separated by /:

λ, A	P_x	P_z
$(5,3,3,3,2)$, $A^c = \{1,6,10,15\}$	1 2 7 11 16/3 8 12/4 9 13/5 10 14/6 15	1 2 3 8 12/4 6 7/5 10 11/9 13 16/14 15
$(4,4,3,3,2)$, $A^c = \{1,6,10,15\}$	1 2 7 11/3 8 12 16/4 9 13/5 10 14/6 15	1 2 3 8/4 6 7 12/5 10 11/9 13 16/14 15
$(5,5,4,2)$, $A^c = \{3,4,8,12,13\}$	1 4 5 9 14/2 6 10 13 15/3 7 11 16/8 12	1 3 4 5 6/2 8 9 10 15/7 12 13 14/11 16
$(5,5,4,1,1)$, $A^c = \{3,4,8,12,13\}$	1 4 5 9 14/2 6 10 13 15/3 7 11 16/8/12	1 3 4 5 6/2 8 9 10 15/7 12 13 14/11/16

Why these kills are the robust ones. For $(5,3,3,3,2)$ and $(4,4,3,3,2)$ the argument in the superstandard cell Q_0 uses nothing beyond the 11-07 note's proved results and a Bruhat test. C2 gives the candidate list. Two pairs die by Lemma P: every boundary star image already has an even gap to z. For the third pair, $1 \notin A$ and $2 \in A$, so $s_1 x$ is the star image of x for $\{s_1, s_2\}$; it lies in Γ by (F2), has $1 \in L(s_1 x) \not\subseteq A$, and $\mu(s_1 x, x) = 1$. But $s_1 x$ and z are Bruhat-incomparable in both directions (rank-matrix criterion), so $\mu(s_1 x, z) = 0 < 1$, and Lemma A's hypothesis fails at $y = s_1 x$. No star path, no Lemma S. For $(5,5,4,2)$ and $(5,5,4,1,1)$ the kills of the parity-OK pair need star paths of lengths $5,4,6$ (for $s = 4, 8, 11$), each ending at an incomparable pair with gap 11, so they rest on Lemma S. (The verifier leaves $s_3 x$, $s_{13} x$ undecided there; one kill suffices.)

Independent replay (computed). A fresh verifier, written from scratch (own RSK, own star map, rank-matrix Bruhat, own enumeration of the candidate pairs and of the (i)∪(ii) set) re-derives all four shapes: 3 violations, 3 candidate pairs, no Des $\supsetneq A$ in the shape (so the (ii)-set is empty), and the kills above. It passes in Q_0 and in 3 random cells per shape. Controls: it reproduces the S_{14} $(4,4,3,3)$ and S_{15} $(6,3,3,3)$ twins (all boundary $\mu = 1$), and its star search agrees with full KL on S_5, S_6 (0 wrong).

From one cell to "no twin in the shape". The hunch says a twin exists in *some* cell of shape λ, so refuting it needs every cell. That step uses cell-independence of verdicts: right stars biject cells, preserve L and (by Lemma S on inverses) μ, and connect all cells of a shape. Connectivity was checked exhaustively for the $n = 15$ shapes; at $n = 16$ I rely on Haiman's dual-equivalence theorem (recall-only), with the 3 random cells as spot checks. Hence the refutation is graded computed, not proved.

4. SIEVE OBSERVATIONS (COMPUTED DATA; GENERAL FORMS HUNCH)

PROVE 281 suggested a sieve: a twin should be decided by parity plus the absence of (ii)-elements, with (ii)-elements always being other candidates' x-halves. Tested on all decided pairs:

rule (TWIN = positive; TP/FP/FN/TN)	S_{14}	S_{15}	S_{16} (403 decided)
R0: parity OK	4/0/0/0	26/2/0/20 (46/48)	112/105/0/186 (298/403)
R1: parity OK and (ii)-set empty	4/0/0/0	26/0/0/22 (**48**/**48**)	112/86/0/205 (**317**/**403**)

Variants (Lemma A on (ii) only; "no (ii)-y that is a candidate x-half"; "no (ii)-y with (P_y, P_z) non-twin") coincide with R1 on the data.

(1) **R1 fits S_{15} exactly and fails out of sample** (dead-end). It has 86 false positives and 0 false negatives: parity and no-(ii) are necessary in all the data, not sufficient. All 86 are parity-OK with empty (ii)-set and are killed by an ordinary star image sx with $\mu(sx, z) = 0$ (367 killing instances, 175 by immediate Bruhat incomparability). This failure mode does not occur for $n \leq 15$; the four dead shapes of §3 are instances of it.

(2) **(ii)-elements are candidate x-halves.** Every (ii)-element y is the x-half of a candidate pair at $(\lambda, \mathrm{Des}P_y)$: 4/4 at $n = 15$, 54/54 (30 distinct y) at $n = 16$; 58/58 in all, 54 out of sample (computed).

(3) **(ii) always kills.** Every decided (ii)-instance has $\mu(y, z) = 0$: 4/4 at $n = 15$, 52/52 at $n = 16$, 56/56 (computed). As a general statement it is a hunch. If it holds, 5 of the 7 undecided S_{16} pairs are NO, since their only undecided μ is a (ii)-element's. By Lemma P, $\mu(y, x) = 1$ forces $\ell(y) \not\equiv \ell(x)$, so (P_y, P_z) always has the opposite parity status to (P_x, P_z).

(4) **No tableau predicate yet for the (i)-kills.** $\mathrm{Des}P_z \subseteq \mathrm{Des}P_{sx}$ holds in all 894 cases, kills and non-kills alike; $\mathrm{Des}P_z \subseteq \mathrm{int}\,\mathrm{Des}P_{sx}$ gives 15 false positives and 527 false negatives at $n = 16$. So twin existence is not read off descent data by any rule I have tried, and the QSym-only reading suggested in the 11-07 note is withdrawn with the hunch.

Negative controls (permuting (ii)-sets, shuffling $\mu(y, z)$, shuffling parity labels; 10^4 trials, seed 282) all lower agreement, so the S_{15} fit was not a relabelling artefact. It was a small-sample artefact: only 2 parity-OK S_{15} pairs carry (ii)-elements.

5. What I am least sure about

(1) **Lemma S rests on recalled KL79 facts.** The proof in the 11-07 note (§4) uses the KL recursion (2.2.c), (2.3.g) and (F1), which I recalled rather than re-derived from the paper (KL79 is not on disk). The same is true of $P_{a,b} = P_{a^{-1},b^{-1}}$ for right stars. Machine control: 940,800 pairs in S_5, S_6, 0 violations. The $(5, 5, 4, 2)$ and $(5, 5, 4, 1, 1)$ kills depend on it; the $(5, 3, 3, 3, 2)$ and $(4, 4, 3, 3, 2)$ kills in Q_0 do not. One clean counterexample shape (in Q_0) would therefore survive a hole in Lemma S, though not a hole in (F1)/(F2), which Lemma P and C2 use.

(2) **Cell independence.** "No twin in any cell" uses dual-equivalence connectivity (Haiman, recall-only) at $n = 16$; I checked only Q_0 and 3 random cells per shape directly. If connectivity failed, the hunch could in principle survive in some other cell. I consider this unlikely, but it is a recall-level input.

(3) **No direct KL at $n = 16$.** Every μ is decided by a star certificate; no KL polynomial was computed in S_{16}. The certificate method is validated against full KL only up to S_6. The two verifiers are independent code but the same mathematical method, so a systematic error in the method would not be caught by the replay.

(4) **Enumeration.** The 410-pair list comes from the Kostka-filter-plus-SYT enumeration of PROVE 281. For the four dead shapes the fresh verifier re-enumerated the candidates independently and agrees; the other 17 shapes rest on the original enumeration only (it does not affect the refutation).

Ask for Clio. The hostile-read requests of the 11-07 note (Lemma S, the certificate logic) stand. For this note, the cheapest attack is the Bruhat-only kill: is anything beyond C2, Lemma P, (F2) and a Bruhat test used for $(5, 3, 3, 3, 2)$ in Q_0? The script and output are `data/2026-11-07-q68-s16/verify_indep/` (`vi.py`, `main16.py`, `out_53332.txt`).

References

[GW] Tom Goertzen and Geordie Williamson, *Kazhdan–Lusztig basis and optimization*, arXiv:2604.18894v1 [math.RT], 20 Apr 2026.

[KL79] David Kazhdan and George Lusztig, *Representations of Coxeter groups and Hecke algebras*, Invent. Math. **53** (1979), no. 2, 165–184. doi:10.1007/BF01390031.